University of Vienna – Business Intelligence 1

Assignment 1 Summer semester 2026: Description

Last updated: March 26, 2026

Goal

The goal of this assignment is for you to go through various facets of business intelligence projects. Starting from the definition of analytical questions you want to answer, through the modeling, transformation, and integration of data, to getting insights to your questions. You can work on any subject of your choice, provided there is suitable data available.

Tasks

You will perform the following tasks:

1. **Topic and data selection:** Think of a domain that interests you for which public data is available. This can be something business-related (e.g. car sales) or non-business-related domains (e.g. grades at a university, COVID infections, chess, movies, etc.). **Find at least two datasets** that relate to this domain and that you can connect to to obtain different kinds of insights. Key aspects to consider here:
    - **You must be able to clearly define what a single entry in your facts table will be**, for instance: a specific sale, the amount of sales of a specific car type in a location per week, etc. Do not continue with the next step if you do not have this settled.
    - **There must be at least 1 measure per fact and at least 3 dimensions surrounding the facts table.** Note: you should almost always have a Date dimension!
    - To find multiple datasets that are compatible, you can look for:
        - datasets that each contain different entries to put in your fact table, e.g., one set with sales in Europe and another with sales in Asia. They must at least have the same granularity level then, though it is acceptable if they have (slightly) different attributes
        - datasets that provide different facts (e.g., different measures) for the same set of entries for your facts table, such as a set of average temperatures per country per month and a set of inflation values per country per month.

    - datasets where one provides information on the facts and another provides additional dimensional information, such as a facts set with data on movie ticket sales and another set with additional information per movie/producer/actor
  - Great sources for finding datasets are https://www.kaggle.com/ and https://datasetsearch.research.google.com/. You can also find your data elsewhere (as long as its public).
  - If you are unsure about the suitability of your data, just ask me.
2. **BI justification and focus:** Place yourself in the position of an analyst tasked to investigate your domain of interest. Establish 3-5 analytical questions that focus on different dimensions (or combinations thereof) that can be answered from your data. Describe what concrete benefits could be gained from finding answers from these questions (recall Session 2 on BI justification).
3. **Data modeling:** This task consists of two parts:
   - **Source data:** Describe and characterize each of your input datasets in a structured manner.
     - Per dataset, provide an ER model in 3NF (third-normal form) to represent it (even if your data is originally a CSV file).
     - For source datasets with lots of columns, you can summarize the attributes (or mention which columns you will not be using later), the focus here should be on which entities your data captures!
     - If the data is unsuitable to be captured in an ER model, justify why that is the case, then describe your data in a different structured manner.
   - **BI data:** Establish a star schema to capture data in a format that will allow you to answer your established analytical questions in a convenient manner. Make sure to:
     - Clearly define what a single entry into your facts table corresponds to
     - Describe the most important design choices you made, e.g., regarding which details to include or which to leave out. These choices need to be tied to your analytical questions.
4. **Data processing: Implement an automatic pipeline** (preferably using Python) to turn your source datasets into data that follows the format of your established star schema. The output of your pipeline should be one or more files that can be loaded into Tableau. This task involves several aspects:

- **Data transformation:** Transform the data into the appropriate structure, deriving relevant attributes (e.g., in a Date dimension), discarding irrelevant data, and establishing relations through primary and foreign keys.
  Note: You do **not** have to first transform your source data into 3NF.
- **Data integration:** Connect your datasets through some shared attributes. Describe which challenges you faced to achieve this integration (e.g., inconsistencies across datasets) and how your script deals with them.
- **Data cleaning:** Assess if your data is subject to quality issues and, where possible, resolve these in an automatic fashion.

Your pipeline should be as reusable as possible, thus try to limit hardcoding or manual cleaning operations. If you need to resort to one of these options, make this explicit and explain why it was necessary.

5. **Data analytics:** Load your data into Tableau and develop an analytical dashboard that, for each of your analytical questions, has one or more components that help to gain the necessary insights. Make sure that your dashboard is well-organized and does not have clutter, such as visualizations that are not relevant to your questions. Use the range of Tableau functionality where possible.
6. **Key insights:** Based on your dashboard, report on the key insights that you obtained for the analytical questions that you established. Think of this task as if you were informing management about your findings.

Groups and Registration

You will work in groups of 4-5 students. Register your group in the appropriate manner on Moodle. You can use the dedicated Moodle forum to look for a group or additional group members. To do not just add yourself to a group of which you don't know the students.

Deliverables and Submission

The project has the following deliverables:

Presentation (May 15, 2026): You will present your work in a 15-minute presentation, followed by a brief Q&A. Some pointers:

- **Coverage:** Your presentation should cover each of the 6 tasks described above, meaning that your presentation will give an overview of the entire flow from Data Selection and BI justification to key insights that you obtained. Some notes:
  - **Focus:** The aim of your presentation is to be appealing and understanding to the audience. Since time is limited, therefore, focus on high-level aspects in your presentation. Technical details (e.g., of Task 4) and other specifics can be put in the additional slides.
  - **Task integration:** Feel free to integrate results of different tasks, for example, the models of your source data sets (Task 3) can be presented at the same time as the statistics of the data (Task 1), whereas the description of the key insights (Task 6) can go hand-in-hand with showing the visualizations of your dashboard (Task 5)
  - **Additional slides:** Additional details per task that you did not cover in your presentation can be put in an appendix, i.e., in additional slides that you did not present but that we can review when assessing your work or that you can show during the Q&A session.
- **Brevity: Time is limited!** Keep the number of slides low and the slides themselves clear and concise. Use appropriate visualizations, focus on concrete bullet-point statements, rather than full paragraphs.
- **Presenters:** It's best to have at most two presenters who present on behalf of the entire group.
- **Attendance:** Attendance is mandatory for the entire group for the entire session, everyone in the group should be able to answer questions regarding the project.

Repository: Your presentation slides should contain a link to repository (e.g., Github/Gitlab) that should contain the following:

- **Data:** Links to your source data and file(s) capturing your transformed data
- **Data processing pipeline:** A Python project, Jupyter Notebook, or equivalent that can be used to automatically transform your source data into the target star schema. In terms of documentation, make sure that it is clear how the pipeline should be executed and what each of the different parts does.
- **Tableau Workbench:** Provide the Tableau workbench you created.

Submission: On the day of the presentation, make sure to submit a PDF of your presentation slides, including optional appendix slides and a link to your project repository via Moodle.

Grading

You can receive a maximum of 35 points for your assignment. Each of the tasks will be assessed in terms of:

- the quality of the solution itself,
- the method used to arrive at that solution, and
- the clarity and quality of the corresponding part in the presentation and additional slides.

You will be graded as a group.

Note: Make sure to carefully read the assignment and ensure that your slide deck (including the additional slides) provides answers to all aspects mentioned in the tasks.

Questions & group issues

For content-related, technical, and administrative questions, please use the Q&A Forum on Moodle so that other groups can also benefit from the answers.

In case of group-related issues, reach out to me via e-mail (han.van.der.aa@univie.ac.at) as soon as possible! I cannot help you if I only hear about issues close to or after the deadline.